Colour-Based Sorting Automation Using a 4-DOF Robotic Arm

19EEE381 – Open Lab

Report

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Program Educational Objectives (PEOs)

PEO1: Graduate can demonstrate electrical and electronics engineering problem solving skill along with proficiency in communication and professional excellence in project management and execution.

PEO2: Graduate can be employable in engineering services including ICT enabled sectors and also motivated for entrepreneurship.

PEO3: Graduate will be competent for higher studies in world class universities and research in industrial organizations.

PEO4: Graduate will manifest social commitment, environmental awareness and moral and ethical values in professional and other discourses.

Program Specific Outcomes (PSOs)

PSO1: Build and manage electro dynamic systems using Knowledge on electrical technology and semiconductor devices for allied services.

PSO2: Use computational tools and network dynamics for design, analysis and control of power systems integrated with renewable energy and Electric Vehicle.

PSO3: Leverage digital technologies employing state-of- the art control techniques and embedded controllers for industrial applications.

BONAFIDE CERTIFICATE

This is to certify that the open lab project report entitled **"Colour-Based Sorting Automation Using a 4-DOF Robotic Arm",**

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is in partial fulfillment of the requirements for the award of the **Degree of Bachelor of Technology** in **"Electrical and Electronics Engineering"** is a bonafide record of the work carried out at Amrita School of Engineering, Coimbatore.

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ABSTRACT

Industrial automation plays a crucial role in enhancing efficiency and precision in manufacturing processes. Conventional sorting mechanisms often rely on manual labor or mechanical systems that lack adaptability. This project focuses on developing a color-based sorting automation system using a 4-DOF robotic arm, ESP32-CAM for color detection, OpenCV for image processing, and an IR sensor for object detection. The system is designed to detect and sort objects of different colors (red, blue, and yellow) into designated bins. The integration of real-time image processing with robotic automation enhances sorting accuracy and efficiency. The ESP32-CAM captures images of incoming objects, processes them using OpenCV to classify colors, and then directs the robotic arm to move the objects to their respective bins. This project aims to minimize human intervention, increase operational speed, and improve sorting accuracy in industrial applications.

INTRODUCTION

In modern industries, automation is essential for improving efficiency, precision, and productivity. Traditional sorting systems either require human intervention or mechanical sorting techniques that lack flexibility and accuracy. Manual sorting is prone to human errors and inefficiencies, while mechanical methods often struggle with varying object characteristics.
Colour-based sorting is widely used in industries such as packaging, food processing, and recycling. A smart, automated sorting system can significantly enhance throughput, reduce errors, and optimize the overall process. The combination of computer vision and robotic automation provides an advanced solution to conventional sorting challenges.

This project aims to develop an intelligent colour-based sorting system using:

- ESP32-CAM for real-time colour detection.
- OpenCV for image processing and classification of objects based on color.
- An IR sensor for object detection and activation of the sorting process.
- A 4-DOF robotic arm for automated sorting and placement of objects into designated bins.

By leveraging these technologies, the system can accurately classify and sort objects based on predefined colour categories, improving efficiency in industrial automation.

PROBLEM STATEMENT

Traditional sorting methods in industries have several limitations:

1. **Manual Sorting Inefficiencies** – Labor-intensive sorting is slow and error-prone.
2. **Fixed Mechanical Sorting** – Conventional mechanical sorting lacks adaptability and precision.
3. **Limited Automation** – Many sorting systems require human supervision, reducing overall efficiency.
4. **Processing Delays** – Inefficient sorting methods increase operational time and costs.

This project aims to address these issues by developing an automated, colour-based sorting system with a robotic arm that ensures precise and efficient sorting, reducing manual effort and improving process reliability.

METHODOLOGY

The implementation follows a structured approach integrating hardware and software components for effective colour-based sorting.

1. **Colour Detection Using ESP32-CAM and OpenCV**
   - ESP32-CAM captures real-time images of objects on the conveyor.
   - OpenCV processes images and classifies objects based on colour (red, blue, yellow).
   - The detected colour is sent as a signal to the robotic arm controller.

2. **Object Detection Using IR Sensor**
   - An IR sensor detects the presence of an object on the conveyor.
   - Upon detection, the camera is triggered to capture an image for colour classification.
   - This ensures that only detected objects are processed, avoiding false triggers.

3. **Automated Sorting with 4-DOF Robotic Arm**
   - The robotic arm receives input from the ESP32-CAM and IR sensor.
   - Based on colour classification, the robotic arm moves the object to the corresponding bin.
   - Servo motors control the movement of the robotic arm for precise positioning.

4. **System Communication and Control**
    - The ESP32-CAM processes image and sends classification results to a microcontroller.
    - The microcontroller controls the robotic arm movements based on the received signals.
    - The system ensures seamless coordination between object detection, colour classification, and sorting actions.

RESULTS

The system was successfully tested in a controlled environment and demonstrated:

- **Accurate Colour Classification** – Objects were correctly identified and sorted into respective bins.
- **Efficient Object Detection** – The IR sensor effectively triggered the sorting mechanism without false detections.
- **Smooth Robotic Arm Operation** – The 4-DOF robotic arm successfully picked and placed objects in designated bins with precision.
- **Real-Time Processing** – The ESP32-CAM and OpenCV-based image processing provided quick and accurate classification.
- **Reduced Human Intervention** – The system functioned autonomously, improving operational efficiency and reducing manual effort.

IMPLEMENTATION CHALLENGES AND SOLUTIONS:

1. **Lighting Conditions Affecting Colour Detection** – Implemented adaptive brightness correction in OpenCV for consistent detection.
2. **IR Sensor Sensitivity Issues** – Optimized sensor positioning to improve object detection accuracy.
3. **Robotic Arm Precision** – Fine-tuned servo motor calibration for smoother and more accurate movements.

CONCLUSIONS & FUTURE WORK

This project successfully demonstrates a colour-based sorting automation system using an ESP32-CAM, OpenCV, an IR sensor, and a 4-DOF robotic arm. The integration of real-time image processing with robotic automation significantly improves sorting accuracy and efficiency, reducing the need for human intervention.
Future enhancements may include:

- **IoT Integration** – Cloud-based monitoring and remote control for enhanced automation.
- **AI-Powered Colour Recognition** – Advanced deep learning models to improve classification accuracy in complex lighting conditions.
- **Multi-Object Sorting** – Expanding the system to handle multiple colours and object types simultaneously.
- **Conveyor Belt Integration** – Automating object transportation for large-scale industrial applications.
- **Optimized Arm Control Algorithms** – Enhancing precision and speed of the robotic arm for faster sorting.

By incorporating these advancements, the system can further improve automation in industrial sorting processes, increasing productivity and reducing operational costs.

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